

Model comparison :

The **Model Comparison Table** is used to show the **performance of all machine learning models together** so it is easy to see **which model performs best**.

Model comparison table :-

Model	Accuracy
Random Forest	0.9990
Gradient Boosting	0.9985
SVM	0.9695
KNN	0.9705

Technical Interpretation

The performance of four machine learning models—Random Forest, Gradient Boosting, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN)—was compared using accuracy as the evaluation metric. Among these models, Random Forest achieved the highest accuracy of **99.90%**, followed by Gradient Boosting with **99.85%**. SVM and KNN showed comparatively lower accuracy.

Business Interpretation

The comparison shows that **Random Forest provides the most reliable predictions for machine failure detection**. This means the model can help maintenance teams identify potential failures early and reduce unexpected machine downtime.

Best Model: Random Forest

Reason: Highest accuracy and better ability to capture complex patterns in machine operational data.